

Analyse IV

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Chapter 1

Review of complex numbers $\mathbb{C}$

1.1 Basic Definitions

A complex number is written as

$$z = x + iy, \quad x = \Re(z) \in \mathbb{R}, \quad y = \Im(z) \in \mathbb{R}, \quad i^2 = -1.$$

Conjugate: $\bar{z} = x - iy$.

Modulus:

$$|z| = \sqrt{x^2 + y^2}.$$

Geometric interpretation: $|z|$ is the distance from z to 0, and $|z - z_0|$ is the distance from z to z_0 .

1.2 Euler's Formula

For $\theta \in \mathbb{R}$,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad \text{thus} \quad |e^{i\theta}| = 1.$$

1.3 Polar Coordinates

If $z \neq 0$, there exist $r = |z| > 0$ and an argument θ such that

$$z = re^{i\theta}, \quad \theta \equiv (z) \pmod{2\pi}.$$

Principal argument: $(z) \in (-\pi, \pi]$.

Products/quotients in polar form: if $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$,

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \quad (z_2 \neq 0).$$

Rotation: multiplying by $e^{i\phi}$ corresponds to a rotation of angle ϕ .

1.4 Algebraic Equations: k -th Roots

To solve $z^k = a$ ($a \neq 0$), write $a = \rho e^{i\alpha}$:

$$z = \rho^{1/k} e^{i(\alpha + 2\pi m)/k}, \quad m = 0, 1, \dots, k-1.$$

Thus there are k **solutions** (counting multiplicities).

Chapter 2

Complex functions

2.1 Complex functions

A function $f : \Omega \subset \mathbb{C} \rightarrow \mathbb{C}$ can be written as

$$f(z) = f(x + iy) = u(x, y) + iv(x, y),$$

where $u, v : \Omega \subset \mathbb{R}^2 \rightarrow \mathbb{R}$.

2.1.1 Complex Exponential and Trigonometry

$$e^{x+iy} = e^x(\cos y + i \sin y).$$

Definitions (via the exponential, consistent with Euler's formula):

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Hyperbolic functions:

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}.$$

2.2 Continuity

f is continuous at z_0 if

$$\forall \varepsilon > 0, \exists \delta > 0 : |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \varepsilon.$$

Equivalent: $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Practical criterion: if $f = u + iv$, then f is continuous at $z_0 = x_0 + iy_0$

$$\iff u \text{ and } v \text{ are continuous at } (x_0, y_0).$$

2.3 Complex Derivative

f is **complex differentiable** at z_0 if the limit exists:

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

2.4 Holomorphic and Entire Functions

Let $\Omega \subset \mathbb{C}$ be **open**.

- f is **holomorphic** on Ω if it is differentiable at every point of Ω .
- f is **entire** if it is holomorphic on $\mathbb{C}$.

Recall: Ω is open if $\forall z_0 \in \Omega, \exists r > 0$ such that the disk $D_r(z_0) = \{z : |z - z_0| < r\} \subset \Omega$.

2.5 Cauchy–Riemann Equations

2.5.1 Theorem (C–R)

Let Ω be open and $f = u + iv$ with $u, v \in C^1(\Omega)$. If f is holomorphic, then

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}} \quad (\text{Cauchy–Riemann equations}).$$

2.5.2 Useful Formula for f'

When C–R are satisfied,

$$\boxed{f'(z) = u_x + iv_x = v_y - iu_y.}$$

2.5.3 Quick Examples

- $f(z) = z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy)$: $u = x^2 - y^2$, $v = 2xy$. We have $u_x = 2x = v_y$ and $u_y = -2y = -v_x$, hence f is holomorphic and $f'(z) = 2z$.
- Conjugation $f(z) = \bar{z} = x - iy$: $u = x$, $v = -y$, so $u_x = 1$ but $v_y = -1 \Rightarrow$ C–R fails: **not holomorphic**.
- $f(z) = e^z = e^x(\cos y + i \sin y)$: C–R holds $\Rightarrow$ holomorphic and $(e^z)' = e^z$.
- $f(z) = \frac{1}{z}$ is holomorphic on $\mathbb{C}^*$ and $f'(z) = -\frac{1}{z^2}$.

2.5.4 Differentiation Rules (same as real case)

If f, g are holomorphic on Ω :

$$(f + g)' = f' + g', \quad (fg)' = f'g + fg', \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0).$$

Chain rule: if $g(\Omega) \subset W$ and f is holomorphic on W , then

$$(f \circ g)'(z) = f'(g(z)) g'(z).$$

Usual consequences:

$$(z^k)' = kz^{k-1}, \quad (\cos z)' = -\sin z, \quad (\sin z)' = \cos z.$$

2.5.5 Geometric Interpretation of C–R (idea)

View $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ via $(x, y) \mapsto (u(x, y), v(x, y))$. If f is holomorphic, the Jacobian matrix Df has the form

$$Df = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix},$$

so locally it is a **scaling** (factor $\sqrt{a^2 + b^2}$) followed by a **rotation** (no anisotropic stretching): small squares remain squares.

2.6 Complex Logarithm

2.6.1 Argument and Discontinuity

(z) (principal value) is defined on $\mathbb{C} \setminus \{0\}$ but is **discontinuous** along the negative real axis (jump from π to $-\pi$).

2.6.2 Definition of the Principal Logarithm

For $z \neq 0$,

$$\boxed{\text{Log}(z) = \log |z| + i(z)}$$

(principal branch).

2.6.3 Properties and Remarks

- Log is not defined at 0 and is not globally continuous (branch cut).
- On a domain where a branch is chosen (e.g. $\mathbb{C} \setminus \{x \leq 0\}$), Log is holomorphic and

$$(\text{Log} z)' = \frac{1}{z}.$$

- Usual rules do not hold *strictly* due to multivaluedness/discontinuity:

$$\text{Log}(z_1 z_2) \neq \text{Log} z_1 + \text{Log} z_2 \quad \text{in general,}$$

but they hold **up to a multiple of $2\pi i$** :

$$\text{Log}(z_1 z_2) = \text{Log} z_1 + \text{Log} z_2 + 2\pi i k, \quad k \in \mathbb{Z}.$$

Chapter 3

Complex integration

Lecture 3 — 10-03-2026: Integration and Cauchy theorem

3.1 Line Integrals in $\mathbb{R}^2$ (Recall)

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a continuous vector field

$$F(x_1, x_2) = (F_1(x_1, x_2), F_2(x_1, x_2))$$

Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be a C^1 curve with image Γ .

The line integral of F along γ is defined as

$$\int_{\Gamma} F \cdot d\ell = \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt$$

which explicitly gives

$$\int_a^b [F_1(\gamma(t))\gamma'_1(t) + F_2(\gamma(t))\gamma'_2(t)] dt$$

Remarks

- The line integral is independent of the parametrization.
- Reversing the orientation changes the sign.
- A curve is called
 - **simple** if it has no self-intersections
 - **regular** if $\gamma'(t) \neq 0$

If Γ is closed and $\nabla \times F = 0$ in the interior of Γ then

$$\int_{\Gamma} F \cdot d\ell = 0$$

(Stokes' theorem).

3.2 Complex Line Integrals

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a regular curve parametrizing $\Gamma \subset \mathbb{C}$.

Let $f : \Gamma \rightarrow \mathbb{C}$ be continuous.

Definition 1. The line integral of f along Γ is

$$\int_{\Gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Orientation

If $-\Gamma$ denotes the curve with opposite orientation then

$$\int_{-\Gamma} f(z) dz = - \int_{\Gamma} f(z) dz$$

Remark

The value of the line integral does not depend on the parametrization as long as the orientation is preserved.

3.3 Examples

3.3.1 Integral of z^2 on the unit circle

Let Γ be the unit circle parametrized by

$$\gamma(t) = e^{it}, \quad t \in [0, 2\pi]$$

Then

$$\gamma'(t) = ie^{it}$$

We compute

$$\begin{aligned} \int_{\Gamma} z^2 dz &= \int_0^{2\pi} (e^{it})^2 (ie^{it}) dt \\ &= i \int_0^{2\pi} e^{3it} dt \\ &= i \left[\frac{e^{3it}}{3i} \right]_0^{2\pi} = 0 \end{aligned}$$

3.3.2 Integral of $\frac{1}{z}$

$$\int_{\Gamma} \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} (ie^{it}) dt = i \int_0^{2\pi} dt = 2\pi i$$

3.3.3 Integral of $\frac{1}{z^2}$

$$\int_{\Gamma} \frac{1}{z^2} dz = \int_0^{2\pi} \frac{1}{e^{2it}} (ie^{it}) dt = i \int_0^{2\pi} e^{-it} dt = 0$$

3.4 Cauchy's Theorem

Theorem 1. *Let $U \subset \mathbb{C}$ be open and simply connected. Let Γ be a closed piecewise regular curve contained in U . If $f : U \rightarrow \mathbb{C}$ is holomorphic then*

$$\int_{\Gamma} f(z) dz = 0$$

Simply connected: the domain has no holes.

Examples:

$$\mathbb{C}, \quad \{z \in \mathbb{C} : |z| < 1\}$$

are simply connected, while

$$\mathbb{C}^* = \mathbb{C} \setminus \{0\}$$

is not.

Piecewise regular: Γ might have finitely many curves, e.g. square.

3.4.1 Idea of the Proof

Write

$$f(z) = u(x, y) + iv(x, y)$$

and parametrize

$$\gamma(t) = \alpha(t) + i\beta(t)$$

Then

$$\int_{\Gamma} f(z) dz = \int_a^b [u(\gamma(t)) + iv(\gamma(t))] (\alpha'(t) + i\beta'(t)) dt$$

Separating real and imaginary parts gives two line integrals in $\mathbb{R}^2$.

Using Green's theorem and the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

both integrals vanish, hence

$$\int_{\Gamma} f(z) dz = 0.$$

3.4.2 Extension of Cauchy's Theorem

Corollary 1. *Let Γ_0 and Γ_1 be closed simple regular curves with*

$$\Gamma_1 \subset \text{int}(\Gamma_0)$$

If f is holomorphic on the region between them then

$$\int_{\Gamma_0} f(z) dz = \int_{\Gamma_1} f(z) dz$$

This allows computation of complicated integrals using simpler curves.

3.5 Cauchy Integral Formula

Theorem 2. *Let $D \subset \mathbb{C}$ be open and $f : D \rightarrow \mathbb{C}$ holomorphic.*

Let Γ be a closed simple curve inside D oriented clockwise and let z_0 lie inside Γ .

Then

$$f(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{z - z_0} dz$$

This formula is particularly useful to compute integrals of the form

$$\int_{\Gamma} \frac{f(z)}{z - z_0} dz.$$

Remark: if the curves is oriented counterclockwise we change the sign as usual.

If more than one z_0 , we can use this formula if only one z_0 is inside Γ , if yes use this formula by using this z_0 .

I.e:

$$I = \int_{\gamma} \frac{e^z}{(z-1)(z^2+4)} dz \quad \gamma = e^{it} \quad t \in [0, 2\pi]$$

Here $z_0 = 2i, -2i$ is out of the unit circle, so only do for $z_0 = 1$

$$f(z) = \frac{e^z}{z^2 + 4}$$

Idea of the proof

Consider a small circle around z_0

$$\gamma_r(t) = z_0 + re^{it}$$

Then

$$\int_{\Gamma} \frac{f(z)}{z - z_0} dz = \int_{\gamma_r} \frac{f(z)}{z - z_0} dz$$

which becomes

$$\int_0^{2\pi} f(z_0 + re^{it}) i r dt$$

As $r \rightarrow 0$, continuity of f gives

$$2\pi i f(z_0)$$

which proves the formula.

Lecture 4 — 12-03-2025: Laurent series and Taylor series

3.5.1 Remark: trick to compute contour integrals

A useful trick to compute contour integrals is the following.

- Find another simple closed curve that surrounds the same point as the complicated curve.
- If the function is holomorphic on the region between the two curves, then the integrals along the two curves are equal.

In other words, if Γ_0 and Γ_1 are two curves with the same endpoints and orientation, and if f is holomorphic on the region between them, then

$$\int_{\Gamma_0} f(z) dz = \int_{\Gamma_1} f(z) dz.$$

The idea is to apply Cauchy's theorem to the closed contour formed by Γ_0 and $-\Gamma_1$.

3.5.2 Extended Cauchy Integral Formula

Under the same assumptions, for every integer $n \geq 0$ we have

Corollary 2.

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\Gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Equivalently,

$$\int_{\Gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0).$$

Remark. This formula shows that holomorphic functions are infinitely differentiable.

3.5.3 Examples

Example 1

Let

$$f(z) = \frac{\cos z}{z}$$

and let Γ be a positively oriented simple closed curve.

We distinguish three cases:

- If 0 lies on Γ , the integral is not defined.
- If 0 is outside Γ , then

$$\int_{\Gamma} \frac{\cos z}{z} dz = 0$$

by Cauchy's theorem.

- If 0 is inside Γ , then using Cauchy's integral formula

$$\int_{\Gamma} \frac{\cos z}{z} dz = 2\pi i \cos(0) = 2\pi i.$$

Example 2

Let $\Gamma = \{z \in \mathbb{C} : |z - 2| = 1\}$ oriented counterclockwise. Compute

$$\int_{\Gamma} \frac{e^z}{(z - 2)^3} dz.$$

Using the extended Cauchy formula with $f(z) = e^z$, $z_0 = 2$ and $n = 2$,

$$\int_{\Gamma} \frac{e^z}{(z - 2)^3} dz = \frac{2\pi i}{2!} f^{(2)}(2).$$

Since

$$f^{(2)}(z_0) = e^2,$$

we obtain

$$\int_{\Gamma} \frac{e^z}{(z - 2)^3} dz = \frac{2\pi i}{2} e^2 = \pi i e^2.$$

Chapter 4

Laurent series and Taylor series

4.1 Taylor Polynomial

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be sufficiently regular (N-times differentiable). For any $x_0 \in \mathbb{R}$, the Taylor polynomial of degree N is

$$T_{x_0}^N f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(N)}(x_0)}{N!}(x - x_0)^N.$$

4.1.1 Taylor's Theorem

If $f \in C^{N+1}$, then for all $x_0, x \in \mathbb{R}$ there exists $\vartheta \in (0, 1)$ such that

$$f(x) = T_{x_0}^N f(x) + \frac{f^{(N+1)}(\vartheta x_0 + (1 - \vartheta)x)}{(N + 1)!}(x - x_0)^{N+1}.$$

Remark. Taylor's theorem shows that a function can be approximated locally by a polynomial.

4.1.2 Taylor Series

If f has derivatives of all orders, we can define the Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

A natural question is whether this series converges and whether it converges to the function f .

4.1.3 Example

For the exponential function,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

This series converges for all $x \in \mathbb{R}$.

Remark: Functions f s.t the Taylor series converges to f in a neighborhood of x_0 are called **analytic**. (Not all smooth (continually derivable) functions are analytic)

4.1.4 Taylor coefficients

Definition 2 (Taylor Coefficients). The Taylor coefficients of a function $f(z)$ around a point z_0 are defined as:

$$c_n = \frac{f^{(n)}(z_0)}{n!}$$

where $f^{(n)}(z_0)$ is the n -th derivative of f evaluated at z_0 .

This comes from assuming f has a power series representation:

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

4.2 Taylor series / regular Laurent series for complex functions

Theorem 3. Let $D \subset \mathbb{C}$ be open and let $z_0 \in D$. Let $R > 0$ such that, the disk $B_R(z_0) = \{z \in \mathbb{C} : |z - z_0| < R\}$ is contained in D .

If $f : D \rightarrow \mathbb{C}$ is holomorphic.

Then for all $z \in B_R(z_0)$

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

The largest possible R is called **radius of convergence** (It's the distance between z_0 and the nearest singularity (not z_0)).

Remark:

- (1) All holo functions are analytic.
- (2) In practice: R is the distance of z_0 from the *nearest* point where f is not holo or not defined.
- (3) We can connect the Taylor series to Cauchy's integral formula

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n = \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

where

$$c_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta$$

for every simple, close curve γ around z_0 with $\text{int } \gamma \subset D$ and counterclockwise oriented.

Caution: The Taylor series representation around a given point is unique! In particular in order to compute the Taylor series at a given point, we can use other "tricks".

Theorem 4 (Laurent series). Let $D \subset \mathbb{C}$ be open and let $z_0 \in D$. Assume $f : D \setminus \{z_0\} \rightarrow \mathbb{C}$ is holomorphic.

Let $R > 0$ such that $B_R(z_0) \subset D$.

Then for all $z \in B_R(z_0) \setminus \{z_0\}$

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n$$

where for every simple closed counterclockwise oriented curve γ such that $\text{int}(\gamma) \subset D$ and z_0 is enclosed by γ ,

$$c_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Remark

- The Laurent series is like a Taylor series but it may include terms with polynomial singularities (negative powers).
- As for Taylor series, the Laurent series representation is unique. Hence if we determine the coefficients c_n by any method (not necessarily by path integrals), we have determined the Laurent series.
- We do not assume that f is holomorphic at z_0 . It might be holomorphic there or not; no hypothesis is made.

Definition 3. Any Laurent series can be split into a **singular part** and a **regular part**:

$$\sum_{n=-\infty}^{\infty} c_n(z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} c_{-n}(z - z_0)^{-n}}_{\text{singular part}} + \underbrace{\sum_{n=0}^{\infty} c_n(z - z_0)^n}_{\text{regular part}}.$$

In other words, a Taylor series is a Laurent series whose singular part vanishes.

Example If $f : D \rightarrow \mathbb{C}$ is holomorphic and $z_0 \in D$, then its Taylor expansion

$$f(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

is a Laurent series whose singular part is zero since for all $n \leq -1$

$$\frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \text{ is holo. near } z_0$$

so by Cauchy's integral formula $\int_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta = 0$

Example For the following function $f : \mathbb{C} \setminus 0 \rightarrow \mathbb{C}$ let find the Laurent series around $z_0 = 0$

1) $f(z) = \frac{1}{z}$, this is already a Laurent series with

$$c_{-1} = 1, \quad c_n = 0 \text{ otherwise.}$$

2) $f(z) = \frac{5z+1}{z^2} = \frac{1}{z^2} + \frac{5}{z} = \sum_{n=-\infty}^{\infty} c_n z^n$ where

$$c_n = \begin{cases} 1 & \text{if } n = -2 \\ 5 & \text{if } n = -1 \\ 0 & \text{otherwise} \end{cases}$$

Remark 1. The Laurent series of a function is unique. Hence, if we compute a power series expansion for f (e.g. from a known expansion), we have determined its Laurent series.

Example

We know:

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \quad |z| < 1$$

Then:

$$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty} (-1)^n z^{2n}$$

So:

$$\frac{1}{1+z^2} = \sum_{k=0}^{\infty} c_k z^k$$

where

$$c_k = \begin{cases} 0 & \text{if } k \text{ odd} \\ (-1)^{k/2} & \text{if } k \text{ even} \end{cases}$$

Example: Laurent expansion around z_0

Let $f : \mathbb{C} \setminus \{0, -1\} \rightarrow \mathbb{C}$

$$f(z) = \frac{1}{z(z+1)}$$

We study the Laurent expansion at z_0 .

Case 1: $z_0 = 0$

$$f(z) = \frac{1}{z} - \frac{1}{z+1}$$

We expand:

$$\frac{1}{z+1} = \sum_{n=0}^{\infty} (-1)^n z^n, \quad |z| < 1$$

Thus:

$$f(z) = \frac{1}{z} - \sum_{n=0}^{\infty} (-1)^n z^n = \sum_{n=-1}^{\infty} (-1)^{n+1} z^n$$

Radius of convergence:

$$R = 1$$

as the nearest singularity is at $z = -1$ (distance of z_0 to the next distinct singular point of f) **Case 2:** $z_0 = -1$

$$f(z) = -\frac{1}{z+1} + \frac{1}{z}$$

Expand around $z = -1$:

$$\frac{1}{z} = \frac{1}{(z+1)-1} = \frac{1}{-1(1-(z+1))} = \frac{-1}{1-(z+1)} = \sum_{n=0}^{\infty} (-1)(z+1)^n, \quad |z+1| < 1$$

So:

$$f(z) = -\frac{1}{z - (-1)} + \sum_{n=0}^{\infty} (-1)(z+1)^n = \sum_{n=1}^{\infty} (-1)(z+1)^n$$

Radius of convergence:

$$R = 1$$

(Here the center = z_0 and the nearest singularity is at $z = 0$ hence the distance is $|0 - (-1)| = 1$)

Case 3: $z_0 \neq 0, -1$

f is holomorphic near z_0 , so f can be written as a regular Taylor series. Converges in any disk around z_0 which does not contain neither 0 and -1.

Radius:

$$R = \min\{|z_0|, |z_0 + 1|\}$$

4.3 Types of Singularities

4.3.1 Regular Point

Definition 4. f has a **regular point** at z_0 if the singular part of its Laurent series at z_0 vanishes.

Remark 2. f is holomorphic at $z_0 \iff z_0$ is a regular point.

z_0 is a regular point $\iff \lim_{z \rightarrow z_0} f(z)$ exists

$$f(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

$z = 0$ is a regular point (no terms of the form $z^{-1}, z^{-2}, \dots$ ($n < 0$))

4.3.2 Pole

Definition 5. f has a **pole of order m** at z_0 if for $n < -m, c_n = 0, c_{-m} \neq 0$:

$$f(z) = \frac{c_{-m}}{(z - z_0)^m} + \dots + \frac{c_{-1}}{z - z_0} + \text{regular terms}$$

$$f(z) = \frac{1}{z} \Rightarrow \text{pole of order 1 at } z = 0$$

$$f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$$

pole of order 2 at $z = -2$, pole of order 1 at $z = 0$.

Remark 3. f has a pole of order m at $z_0 \iff \lim_{z \rightarrow z_0} (z - z_0)^m f(z)$ exist and is not zero.

In particular if f has a pole of order m at z_0 we can write

$$f(z) = \frac{g(z)}{(z - z_0)^m}$$

where g is holo. and $g(z_0) \neq 0$ (opposite is also true).

4.3.3 Essential Singularity

Definition 6. f has an **essential singularity** at z_0 if its Laurent series has infinitely many coefficient $c_n \neq 0$ with $n \leq -1$.

$$f(z) = e^{1/z} \text{ at } z_0 = 0$$

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = \sum_{n=-\infty}^0 \frac{1}{(-n)!} z^n$$

Essential singularity at $z = 0$.

4.4 Quotients of Taylor Series

Let $p, q : D \setminus \{z_0\} \rightarrow \mathbb{C}$ holo with z_0 being regular for both. Write their Laurent series as:

$$p(z) = a_0 + a_1(z - z_0) + \dots + a_k(z - z_0)^k \quad q(z) = b_0 + b_1(z - z_0) + \dots + b_\ell(z - z_0)^\ell$$

Consider:

$$f(z) = \frac{p(z)}{q(z)}$$

if $b_0 \neq 0$ so $q(z_0) \neq 0$. We have

$$\lim_{z \rightarrow z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0} \text{ exists, so } z_0 \text{ is a regular point for } \frac{p}{q}$$

if $a_0 = b_0$ but $b_1 \neq 0$ then we have

$$\lim_{z \rightarrow z_0} \frac{a_1(z - z_0) + a_2(z - z_0)^2 + \dots}{b_1(z - z_0) + b_2(z - z_0)^2 + \dots} = \lim_{z \rightarrow z_0} \frac{a_1 + a_2(z - z_0) + \dots}{b_1 + b_2(z - z_0) + \dots} = \frac{a_1}{b_1}$$

exists hence z_0 is a regular point for f . (similarly if $a_0 = b_0 = 0$ and $a_1 = b_1 = 0$ yet $b_2 \neq 0$)

In summary, either expand in power series and take out as many common powers of $z - z_0$ from the nominator and denominator, and take the **limit** or us **l'Hôpital's rule**

- If $k \geq \ell$: regular point, as the limit $= \begin{cases} 0 & \text{if } k > \ell \\ \frac{a_k}{b_\ell} & \text{if } k = \ell \end{cases}$ exists
- If $k < \ell$: pole of order $\ell - k$

Example

$$f(z) = \frac{z}{\sin z}$$

where $z_0 = 0$

Using:

$$\sin z = z - \frac{z^3}{3!} + \dots$$

We get:

$$f(z) = \frac{z}{z(1 - z^2/3! + \dots)} = \frac{1}{1 - z^2/6 + \dots}$$

Hence $\lim_{z \rightarrow z_0} = 1$ exist, so $z = 0$ is a regular point.

Example

$$f(z) = \frac{\sin z}{2z^3 + z^4}$$

Using expansions:

$$\sin z = z - \frac{z^3}{6} + \cdots$$

We get:

$$f(z) = \frac{1}{z^2} \cdot (\text{holomorphic nonzero function})$$

So pole of order 2 at $z = 0$.

Lecture 6 — 26-03-2026: Residues and Residue Theorem

4.5 Residues

Definition 7 (Residues). Let f be holomorphic on a punctured neighborhood of z_0 , and write its Laurent series at z_0 as

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n.$$

The coefficient c_{-1} of the term $(z - z_0)^{-1}$ is called the *residue* of f at z_0 :

$$\text{Res}_{z_0}(f) = c_{-1}$$

4.5.1 Integral expression of the residue

By the definition of the coefficients in the Laurent series,

$$\text{Res}_{z_0}(f) = \frac{1}{2\pi i} \int_{\gamma} f(z) dz,$$

where γ is a closed, simple, counterclockwise oriented curve that contains z_0 in its interior, and such that f is holomorphic on $\text{int}_{\gamma} \setminus \{z_0\}$.

Example If f is holomorphic on an open set containing z_0 , then

$$\text{Res}_{z_0}(f) = 0$$

by Cauchy's theorem.

For

$$f(z) = \frac{1}{z},$$

since it's already the Laurent series at 0, hence

$$\text{Res}_0\left(\frac{1}{z}\right) = 1$$

4.5.2 Formula for a pole of order m

Suppose that f has a pole of order m at z_0 , so that its Laurent series is

$$f(z) = c_{-m}(z - z_0)^{-m} + \cdots + c_{-1}(z - z_0)^{-1} + c_0 + c_1(z - z_0) + \cdots$$

Consider the holomorphic function

$$g(z) = (z - z_0)^m f(z) = c_{-m} + c_{-m+1}(z - z_0) + \cdots + c_{-1}(z - z_0)^{m-1} + c_0(z - z_0)^m + \cdots$$

Therefore,

Corollary 3.

$$c_{-1} = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left((z - z_0)^m f(z) \right)$$

This formula is useful when we already know that f has a pole of order m at z_0 .

In particular, if $m = 1$, then

$$\operatorname{Res}_{z_0}(f) = \lim_{z \rightarrow z_0} (z - z_0) f(z).$$

Example

$$f(z) = \frac{1}{\sin z}$$

At $z_0 = 0$, this function has a pole of order 1.

Hence,

$$\operatorname{Res}_0(f) = \lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} \frac{z}{\sin z}$$

Using

$$\sin z = z - \frac{z^3}{3!} + \cdots,$$

we get

$$\frac{z}{\sin z} = \frac{z}{z - \frac{z^3}{3!} + \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \cdots},$$

so the limit is

$$\operatorname{Res}_0(f) = 1.$$

4.6 Residue theorem

Theorem 5 (Residue theorem). *Suppose $D \subseteq \mathbb{C}$ is open and simply connected, and let γ be a simple closed curve in D which is counterclockwise oriented. Let $z_1, \dots, z_m \in \operatorname{int}(\gamma)$. Assume*

$$f : D \setminus \{z_1, \dots, z_m\} \rightarrow \mathbb{C}$$

is holomorphic. Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{i=1}^m \operatorname{Res}_{z_i}(f)$$

Hence, in order to compute $\int_{\gamma} f(z) dz$, it suffices to compute the residues of the singular points in the interior of γ .

4.6.1 Idea of the proof

For simplicity, assume there are only two singular points z_1 and z_2 . One deforms the contour γ into two small positively oriented curves γ_1 and γ_2 around the singularities, connected by a very narrow bridge. By Cauchy's theorem, the integral over γ is equal to the sum of the integrals over the small circles, since the contributions of the bridge cancel. Therefore,

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz = 2\pi i \operatorname{Res}_{z_1}(f) + 2\pi i \operatorname{Res}_{z_2}(f).$$

Example

$$f(z) = \frac{1}{z}.$$

Let γ be a simple closed counterclockwise oriented curve.

1. If $0 \in \operatorname{int}(\gamma)$, then by the residue theorem,

$$\int_{\gamma} \frac{1}{z} dz = 2\pi i \operatorname{Res}_0\left(\frac{1}{z}\right) = 2\pi i.$$

2. If $0 \notin \operatorname{int}(\gamma)$, then

$$\int_{\gamma} \frac{1}{z} dz = 0.$$

3. If $0 \in \gamma$, then the integral is not well-defined.

Example

$$f(z) = \frac{1}{z(z+1)}.$$

We have two simple poles, at $z = 0$ and $z = -1$.

Their residues are

$$\operatorname{Res}_0(f) = \lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} \frac{1}{z+1} = 1,$$

and

$$\operatorname{Res}_{-1}(f) = \lim_{z \rightarrow -1} (z+1) f(z) = \lim_{z \rightarrow -1} \frac{1}{z} = -1.$$

So:

1. If neither 0 nor -1 lies in $\operatorname{int}(\gamma)$, then

$$\int_{\gamma} f(z) dz = 0.$$

2. If $0 \in \operatorname{int}(\gamma)$ but $-1 \notin \operatorname{int}(\gamma)$, then

$$\int_{\gamma} f(z) dz = 2\pi i.$$

3. If $-1 \in \operatorname{int}(\gamma)$ but $0 \notin \operatorname{int}(\gamma)$, then

$$\int_{\gamma} f(z) dz = -2\pi i.$$

4. If both 0 and -1 lie in $\operatorname{int}(\gamma)$, then

$$\int_{\gamma} f(z) dz = 2\pi i (\operatorname{Res}_0(f) + \operatorname{Res}_{-1}(f)) = 2\pi i (1 - 1) = 0.$$

5. If $0 \in \gamma$ or $-1 \in \gamma$, then the integral is not well-defined.

Example $f : \mathbb{C} \setminus \{0, 1\} \rightarrow \mathbb{C}$,

$$f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}.$$

Then f has a pole of order 2 at 0 and a pole of order 1 at 1.

At $z = 1$, since the pole is simple,

$$\text{Res}_1(f) = \lim_{z \rightarrow 1} (z-1)f(z) = \lim_{z \rightarrow 1} \left(\frac{z-1}{z^2} + \frac{2(z-1)}{z} + 3 \right) = 3.$$

At $z = 0$, we use the formula for a pole of order 2:

$$\text{Res}_0(f) = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} (z^2 f(z)).$$

Now

$$z^2 f(z) = 1 + 2z + \frac{3z^2}{z-1},$$

so

$$\frac{d}{dz} (z^2 f(z)) = 2 + \frac{6z}{z-1} - \frac{3z^2}{(z-1)^2}.$$

Evaluating at $z = 0$ gives

$$\text{Res}_0(f) = 2.$$

Actually this could have been seen by reading directly $f(z)$.

If $\gamma(t) = 10e^{it}$ for $t \in [0, 2\pi]$, then both poles lie inside γ , hence

$$\int_{\gamma} f(z) dz = 2\pi i (\text{Res}_0(f) + \text{Res}_1(f)) = 2\pi i (2 + 3) = 10\pi i.$$

4.6.2 Useful remark

It is often advisable to first rewrite the function in a form where the residue can be read off immediately.

For example:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \cdots \implies \text{Res}_0(e^{1/z}) = 1.$$

On the other hand,

$$e^{1/z^2} = 1 + \frac{1}{z^2} + \frac{1}{2!} \frac{1}{z^4} + \cdots \implies \text{Res}_0(e^{1/z^2}) = 0.$$

4.7 Applications of Residues to Real Integrals

4.7.1 Goal

Use complex analysis to compute real integrals of the form:

$$\int_{-\infty}^{\infty} R(x) dx \quad \text{or} \quad \int_{-\infty}^{\infty} R(x) \cos(ax) dx$$

4.7.2 Contour Method

For integral of the form:

$$\int_{-\infty}^{\infty} R(x) dx$$

with

$$R(x) = \frac{P(x)}{Q(x)}$$

Assume:

- $\deg Q \geq \deg P + 2$
- $Q(x) \neq 0$ for all $x \in \mathbb{R}$

$$R(x) = f(x)$$

The idea is to compute $\int_{-\infty}^{\infty} f(x) dx$ by embedding the real line into $\mathbb{C}$ and exploiting the residue theorem.

Step 1 — Build a closed contour. Fix $R > 0$ and define two pieces:

- The **real segment** $[-R, R]$, traversed from left to right.
- The **semicircular arc** $C_R = \{Re^{it} : t \in [0, \pi]\}$, traversed counterclockwise in the upper half-plane.

Together they form a closed contour γ_R , so by the residue theorem:

$$\int_{\gamma_R} f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz = 2\pi i \sum_{\substack{z_k \text{ pole of } f \\ \text{Im}(z_k) > 0}} \text{Res}_{z_k}(f).$$

Step 2 — Show the arc vanishes. One needs to verify that $\int_{C_R} f(z) dz \rightarrow 0$ as $R \rightarrow \infty$. If this holds, taking $R \rightarrow \infty$ gives

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\substack{z_k \text{ pole of } f \\ \text{Im}(z_k) > 0}} \text{Res}_{z_k}(f).$$

Remark 4 (Why the arc vanishes — the key estimate). On C_R we have $|z| = R$, so if $f(z) = P(z)/Q(z)$ with $\deg Q \geq \deg P + 2$, then $|f(z)| \leq C/R^2$. Since the length of C_R is πR , we get

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{C}{R^2} \cdot \pi R = \frac{\pi C}{R} \xrightarrow{R \rightarrow \infty} 0.$$

This is the standard sufficient condition.

Remark 5 (Choice of half-plane for e^{iaz}). When $f(z) = R(z)e^{iaz}$, the exponential factor requires

care. For $z = x + iy$:

$$|e^{iaz}| = e^{-ay}.$$

- If $a > 0$: this decays when $y > 0 \Rightarrow$ use the **upper** half-plane.
- If $a < 0$: this decays when $y < 0 \Rightarrow$ use the **lower** half-plane.
- If $a = 0$: both choices work.

One must choose the half-plane *after* looking at a , not before. This is why such integrals are treated case by case in a .

Example Compute $I = \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Setup. Consider $f(z) = \frac{1}{1+z^2} = \frac{1}{(z-i)(z+i)}$. The poles are at $z = i$ (upper half-plane) and $z = -i$ (lower half-plane).

Contour. We close in the upper half-plane with $\gamma_R = [-R, R] \cup C_R$.

Arc estimate. On C_R , $|f(z)| = \frac{1}{|1+z^2|} \leq \frac{1}{R^2-1}$, so

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{\pi R}{R^2-1} \xrightarrow{R \rightarrow \infty} 0. \checkmark$$

Residue. Only $z = i$ lies in the upper half-plane:

$$\text{Res}_i(f) = \lim_{z \rightarrow i} (z-i)f(z) = \frac{1}{i-(-i)} = \frac{1}{2i}.$$

Conclusion.

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 2\pi i \cdot \frac{1}{2i} = \pi.$$

(Consistent with $\arctan(x)|_{-\infty}^{+\infty} = \pi$.)

4.7.3 Exponential Trick (Euler)

Using:

$$e^{iax} = \cos(ax) + i \sin(ax)$$

We compute:

$$\int_{-\infty}^{\infty} R(x) e^{iax} dx$$

and extract real or imaginary parts.

Example We want to compute

$$\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} \cos x \, dx.$$

This is of the form of the previous example with $a = 1$, $P(x) = x^2$, $Q(x) = 16 + x^4$. Thus,

$$\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} \cos x \, dx = \Re \left(\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} e^{ix} \, dx \right)$$

by Euler's formula.

Define

$$f(z) = \frac{z^2}{16+z^4} e^{iz}.$$

The singularities correspond to the complex roots of

$$16 + z^4 = 0.$$

Solve:

$$z^4 = -16 = 16e^{i(\pi+2k\pi)}, \quad k \in \{0, 1, 2, 3\}.$$

Thus

$$z = 2e^{i\left(\frac{\pi}{4} + \frac{k\pi}{2}\right)}, \quad k = 0, 1, 2, 3.$$

Explicitly:

$$\begin{aligned} z_1 &= \sqrt{2} + i\sqrt{2}, & z_2 &= -\sqrt{2} + i\sqrt{2}, \\ z_3 &= -\sqrt{2} - i\sqrt{2}, & z_4 &= \sqrt{2} - i\sqrt{2}. \end{aligned}$$

Hence

$$Q(z) = (z - z_1)(z - z_2)(z - z_3)(z - z_4),$$

so

$$R(z) = \frac{z^2}{(z - z_1)(z - z_2)(z - z_3)(z - z_4)}.$$

Each z_k is a simple pole.

Only the poles in the upper half-plane contribute:

$$z_1, z_2 \quad (\text{since } \Im z_1, \Im z_2 \geq 0).$$

Compute residues:

$$\text{Res}_{z_1}(f) = \lim_{z \rightarrow z_1} (z - z_1)f(z) = \frac{z_1^2 e^{iz_1}}{(z_1 - z_2)(z_1 - z_3)(z_1 - z_4)}.$$

$$\text{Res}_{z_2}(f) = \lim_{z \rightarrow z_2} (z - z_2)f(z) = \frac{z_2^2 e^{iz_2}}{(z_2 - z_1)(z_2 - z_3)(z_2 - z_4)}.$$

After simplification (as in the notes), we obtain:

$$\text{Res}_{z_1}(f) = \frac{e^{-\sqrt{2}} e^{i\sqrt{2}}}{4\sqrt{2}} (1 + i),$$

$$\text{Res}_{z_2}(f) = \frac{e^{-\sqrt{2}} e^{-i\sqrt{2}}}{4\sqrt{2}} (1 - i).$$

Thus:

$$\Re \left(\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} e^{ix} \, dx \right) = \Re (2\pi i (\text{Res}_{z_1}(f) + \text{Res}_{z_2}(f))).$$

Final result:

$$\int_{-\infty}^{\infty} \frac{x^2}{16+x^4} \cos x \, dx = 4\pi e^{-\sqrt{2}} (\cos(\sqrt{2}) - \sin(\sqrt{2})).$$

4.7.4 Trigonometric Integrals

Using substitution:

$$z = e^{it}$$

we transform:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} dt$$

into a contour integral on the unit circle:

$$\oint f(z) dz$$

4.7.5 Meromorphic Functions

A function holomorphic except at isolated poles is called **meromorphic**.

Residue theorem applies directly to such functions.

4.7.6 Global Strategy

- Extend real integral to complex plane
- Choose appropriate contour
- Show arc integral $\rightarrow 0$
- Identify poles in upper half-plane
- Compute residues
- Apply residue theorem

Chapter 5

Fourier Analysis

5.1 Review of the Fourier Transform

Definition 8 (Fourier Transform). The **Fourier transform** $\hat{f} : \mathbb{R} \rightarrow \mathbb{C}$ of $f : \mathbb{R} \rightarrow \mathbb{C}$, whenever well-defined, is

$$\hat{f}(a) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x) dx.$$

Sometimes written $\mathcal{F}[f]$ instead of $\hat{f}$. It is well-defined if f is piecewise continuous and $\int_{-\infty}^{\infty} |f(x)| dx < \infty$ (symbolically, $f \in L^1(\mathbb{R}; \mathbb{C})$).

Definition 9 (Inverse Fourier Transform). The **inverse Fourier transform** $\check{f} : \mathbb{R} \rightarrow \mathbb{C}$ of $\hat{f} : \mathbb{R} \rightarrow \mathbb{C}$, whenever well-defined, is

$$\check{f}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iax} \hat{f}(a) da.$$

Sometimes written $\mathcal{F}^{-1}[\hat{f}]$ instead of $\check{f}$. Under appropriate hypotheses on f ,

$$\mathcal{F}^{-1}[\mathcal{F}[f]] = f.$$

Remark 6. In general $\hat{f}$ is complex-valued even if f is real-valued. However, if f is real-valued and even (i.e. $f(x) = f(-x)$ for all $x \in \mathbb{R}$), then $\hat{f}$ is also real-valued. Intuitively, one can think of f as a signal in the *physical domain* and $\hat{f}$ as its representation in the *frequency domain*: $\hat{f}(a)$ encodes “how much of frequency a is present in the signal f ”.

5.2 Properties of the Fourier Transform

(1) Linearity

$$\mathcal{F}[\lambda f + \mu g] = \lambda \mathcal{F}[f] + \mu \mathcal{F}[g]$$

whenever $\lambda, \mu \in \mathbb{R}$.

(2) Time Shift Let $g(x) = f(x - x_0)$. Then

$$\hat{g}(a) = e^{-iax_0} \hat{f}(a).$$

Computation.

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x - x_0) dx \stackrel{y=x-x_0}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia(y+x_0)} f(y) dy = e^{-iax_0} \hat{f}(a).$$

□

(3) Frequency Shift Let $g(x) = e^{ia_0x} f(x)$ for $a_0 \in \mathbb{R}$. Then

$$\hat{g}(a) = \hat{f}(a - a_0).$$

This is the “inversion” of property (2), as expected.

(4) Rescaling / Dilation Let $g(x) = f(\lambda x)$ where $\lambda \in \mathbb{R} \setminus \{0\}$. Then

$$\hat{g}(a) = \frac{1}{|\lambda|} \hat{f}\left(\frac{a}{\lambda}\right).$$

Computation. Set $y = \lambda x$ (so $dy = \lambda dx$). Two cases arise:

- If $\lambda > 0$: $\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy$.
- If $\lambda < 0$: $\int_{\infty}^{-\infty} \cdots \frac{1}{\lambda} dy = -\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy = \int_{-\infty}^{\infty} \cdots \frac{1}{|\lambda|} dy$.

In both cases,

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ia\frac{y}{\lambda}} f(y) \frac{dy}{|\lambda|} = \frac{1}{|\lambda|} \hat{f}\left(\frac{a}{\lambda}\right).$$

□

Remark 7 (Uncertainty Principle). Pictorially, as $\lambda \rightarrow \infty$ the graph of $f(\lambda x)$ becomes very concentrated (width $\sim \frac{1}{\lambda}$), while $\frac{1}{|\lambda|} \hat{f}(a/\lambda)$ becomes very spread out. Hence, when f is “very concentrated”, $\hat{f}$ is very spread out and vice versa. This is a manifestation of the **uncertainty principle**.

(5) Derivatives Let $g(x) = f'(x)$. Then

$$\hat{g}(a) = ia \hat{f}(a).$$

More generally, for all $n \in \mathbb{N}$,

$$\mathcal{F}[f^{(n)}](a) = (ia)^n \mathcal{F}[f](a).$$

So the Fourier transform converts differential equations with constant coefficients into *algebraic* equations on the frequency domain.

(6) Convolution Given $f, g : \mathbb{R} \rightarrow \mathbb{C}$, the **convolution** $f * g : \mathbb{R} \rightarrow \mathbb{C}$, whenever well-defined, is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy.$$

Some properties of convolution:

- $f * g = g * f$ (symmetry)
- $f * (g * h) = (f * g) * h$ (associativity)
- $f * (g + h) = f * g + f * h$ (distributivity)
- $(f * g)' = f' * g = f * g'$, so $f * g$ is at least as smooth as the smoother function among f and g .

The convolution can be interpreted as the “local average” of f with weight g .

Example If $g(x) = \frac{1}{2\varepsilon} \mathbf{1}_{[-\varepsilon, \varepsilon]}(x)$, then

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(y) dy,$$

By putting $x - y = -\varepsilon$ and $x - y = \varepsilon$.

which is the local average of f around x . As $\varepsilon \rightarrow 0$, $(f * g)(x) \rightarrow f(x)$ whenever f is continuous at x .

The Fourier transform converts convolutions into products:

$$\mathcal{F}[f * g] = \sqrt{2\pi} \mathcal{F}[f] \cdot \mathcal{F}[g].$$

In applications, one can often express $\mathcal{F}[f]$ as a product of known Fourier transforms, which allows one to recover f by convolution.

(7) Gaussian Function If $f(x) = e^{-x^2/2}$, then

$$\hat{f}(a) = e^{-a^2/2}.$$

(8) Plancherel’s Theorem

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(a)|^2 da.$$

This expresses that f and $\hat{f}$ carry the same “energy”.

5.3 Computing Fourier Transforms via Complex Analysis

Complex analysis, and in particular the residue theorem, provides a powerful method for computing Fourier transforms explicitly.

Example Let $f(x) = \frac{1}{1+x^2}$. Then

$$\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{-iax}}{1+x^2} dx.$$

We compute this using the residue theorem. Consider $a \leq 0$ and set

$$f(z) = \frac{e^{-iaz}}{1+z^2} = \frac{e^{-iaz}}{(z-i)(z+i)}.$$

Since $a \leq 0$, for $z = x + iy$ with $\text{Im}(z) = y > 0$ we have $|e^{-iaz}| = e^{-a(-y)} = e^{ay} \cdot e^{???} \dots$ more precisely, $e^{-iaz} = e^{-ia(x+iy)} = e^{-iax} e^{ay}$. Since $a \leq 0$ and $y > 0$, e^{ay} decays exponentially as $R \rightarrow \infty$. Hence the integral over the semicircle in the upper half-plane vanishes.

The only pole in the upper half-plane is at $z = i$. Using the residue theorem:

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \operatorname{Res}_i(f).$$

Since i is a simple pole:

$$\operatorname{Res}_i(f) = \lim_{z \rightarrow i} (z - i) f(z) = \frac{e^{-ia \cdot i}}{2i} = \frac{e^a}{2i}.$$

Therefore

$$\int_{-\infty}^{\infty} \frac{e^{-iax}}{1+x^2} dx = 2\pi i \cdot \frac{e^a}{2i} = \pi e^a.$$

So for $a \leq 0$:

$$\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \frac{1}{\sqrt{2\pi}} \cdot \pi e^a = \sqrt{\frac{\pi}{2}} e^a.$$

A similar calculation for $a > 0$ using the semicircle in the lower half-plane gives $\sqrt{\frac{\pi}{2}} e^{-a}$. Combining both cases:

$$\boxed{\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \sqrt{\frac{\pi}{2}} e^{-|a|}.}$$

Chapter 6

Laplace Analysis

6.1 The Laplace Transform

Comparison with the Fourier transform:

- Fourier transform: computed for signals defined on $\mathbb{R}$.
- Laplace transform: computed for signals defined on $\mathbb{R}_+ = [0, \infty)$.

Both transforms convert differential equations into algebraic equations. The Laplace transform is a “generalization” in the sense that there are many functions which are not Fourier-transformable but are Laplace-transformable.

Definition 10 (Laplace Transform). Let $f : \mathbb{R}_+ \rightarrow \mathbb{C}$. The **Laplace transform** of f at $z \in \mathbb{C}$, whenever well-defined, is

$$\mathcal{L}[f](z) := \int_0^\infty f(t) e^{-zt} dt.$$

Sometimes the Laplace transform of f is written by the corresponding capital letter, e.g. $F(z) = \mathcal{L}[f](z)$.

6.2 Convergence Criteria

Definition 11 (Abcissa of Convergence). A value $\gamma_0 \in \mathbb{R}$ such that $\mathcal{L}[f](z)$ is well-defined at least on $\{\operatorname{Re}(z) > \gamma_0\}$ is called an **abcissa of convergence**.

The integral $\mathcal{L}[f](z)$ is well-defined if $\int_0^\infty |f(t)| e^{-\operatorname{Re}(z)t} dt < \infty$.

Theorem 6 (Convergence Criterion). Assume there exists $\gamma_0 \in \mathbb{R}$ such that

$$\int_0^\infty |f(t)| e^{-\gamma_0 t} dt < \infty$$

(intuitively, this means $|f(t)| < e^{\gamma_0 t}$ as $t \rightarrow \infty$). Then for $\operatorname{Re}(z) \geq \gamma_0$:

$$|\mathcal{L}[f](z)| = \int_0^\infty |f(t)| |e^{-\operatorname{Re}(z)t}| |e^{-\operatorname{Im}(z)ti}| dt = \int_0^\infty |f(t)| e^{-\operatorname{Re}(z)t} dt \leq \int_0^\infty |f(t)| e^{-\gamma_0 t} dt < \infty.$$

Hence $\mathcal{L}[f](z)$ is defined at least for $\{z \in \mathbb{C} : \operatorname{Re}(z) > \gamma_0\}$.

Example If $f(t) = 0$ for $t \geq M$ and f is piecewise continuous, then

$$\int_0^\infty |f(t)| e^{-\gamma_0 t} dt = \int_0^M |f(t)| e^{-\gamma_0 t} dt < \infty$$

for all $\gamma_0 \in \mathbb{R}$. In this case $\mathcal{L}[f](z)$ is defined for all $z \in \mathbb{C}$.

Remark 8. Let γ_0 be an abscissa of convergence. Then $F(z) = \mathcal{L}[f](z)$ is **holomorphic** on $\{\operatorname{Re}(z) > \gamma_0\}$, and

$$F'(z) = \frac{d}{dz} \int_0^\infty f(t) e^{-zt} dt = \int_0^\infty f(t) (-t) e^{-zt} dt = -\mathcal{L}[tf(t)](z).$$

In short:

$$\boxed{\mathcal{L}[f(t)]' = -\mathcal{L}[tf(t)].}$$

(Here it is implicitly used that if $\gamma_0 \in \mathbb{R}$ is an abscissa of convergence for $\mathcal{L}[f(t)]$, it is also one for $\mathcal{L}[tf(t)]$.)

Example Let $f(t) = 1$. Then for every $\gamma_0 > 0$,

$$\int_0^\infty |f(t)| e^{-\gamma_0 t} dt = \int_0^\infty e^{-\gamma_0 t} dt = \frac{1}{\gamma_0} < \infty,$$

so any $\gamma_0 > 0$ is an abscissa of convergence, and $\mathcal{L}[f](z)$ is defined for $z \in \{\operatorname{Re}(z) > 0\}$. Moreover,

$$\mathcal{L}[1](z) = \int_0^\infty e^{-zt} dt = \left[\frac{e^{-zt}}{-z} \right]_0^\infty.$$

Since $\operatorname{Re}(z) > 0$ we have $\lim_{t \rightarrow \infty} e^{-zt} = 0$ (

because $|e^{-zt}| = |e^{\operatorname{Re} z t}| |e^{-\operatorname{Im} z t i}|$ (real part $\rightarrow 0$ and $\operatorname{Im} = 1$ as $t \rightarrow \infty$)

$$\boxed{\mathcal{L}[1](z) = \frac{1}{z}, \quad \operatorname{Re}(z) > 0.}$$